

Who Should Enter? Reputation, Risk, and Decision Making for UAVs in Low-Altitude Airspace

Grant #: 80NSSC23M0058 | September 15, 2026



SPEAKERS

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Usman Gohar (Iowa State University)

Robert Chris Roberts (University of Texas
El Paso)

Ankit Agrawal (Northeastern Illinois
University)

Lilly Spirkovska (NASA Point of Contact)



Our Team

2



Research Team Members:



Jane
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Yanchen
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Siddharth
Joshi



Tareq
Mahmoud



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Chambers



Kshama Lakshmi
Ranganatha



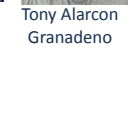
Arturo
Russell



Jonathan Granda
Acaro



Demetrius
Hernandez



Tony Alarcon
Granadeno



Lax
Vadnala



Myra B. Cohen



Robyn Lutz



Usman Gohar



Michael Hunter



Salil Purandare



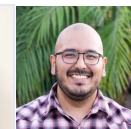
Jordan Rios



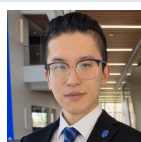
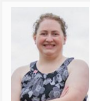
Andrew De Gala



Andrew Soupir

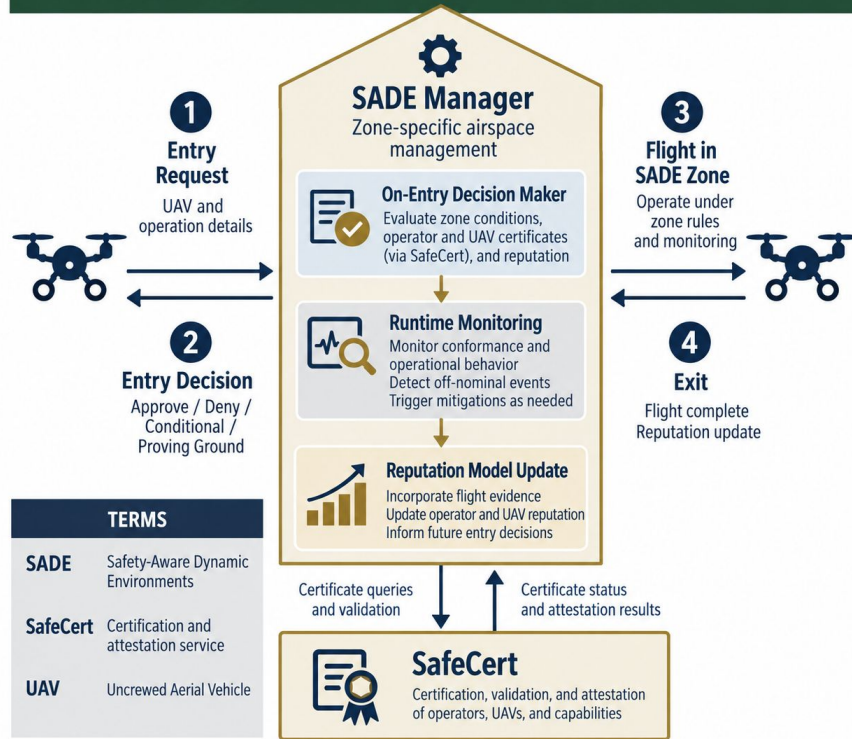


Advisory Committee



SADE Zone

Dynamic, context-specific rules and risk management



Challenge:

- Safe flights of sUAS in managed air spaces with privacy-preserving, fair & **automated** decision making.

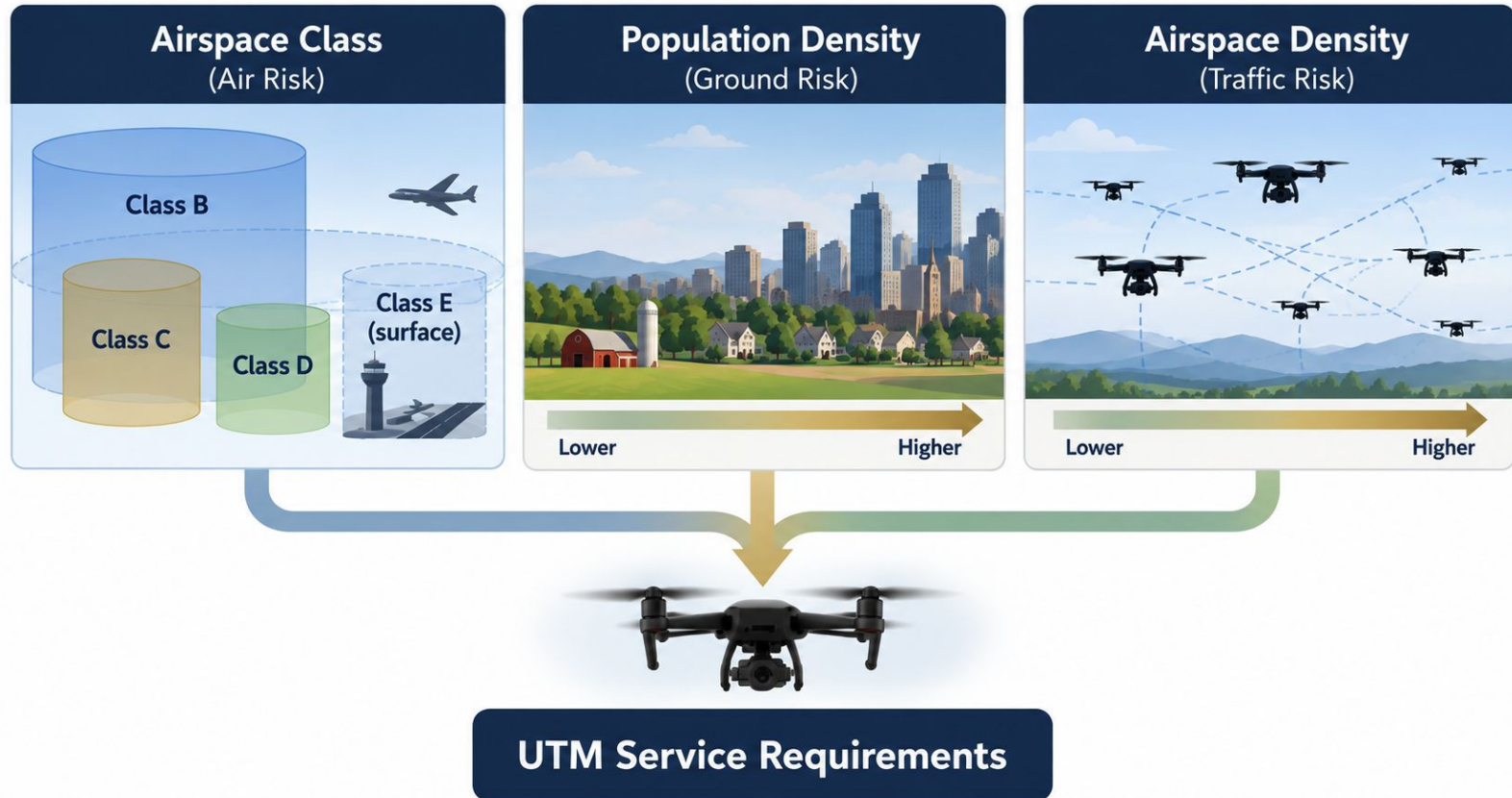
Solution:

- Automate the on-entry decision-making process, through **fair**, **transparent**, **explainable** and **consistent** decisions.
- Supported by onboard analytics to capture flight behavior in privacy preserving, queryable, reputation models; reliable communication via chirper, and high-fidelity, scalable, simulation environments.

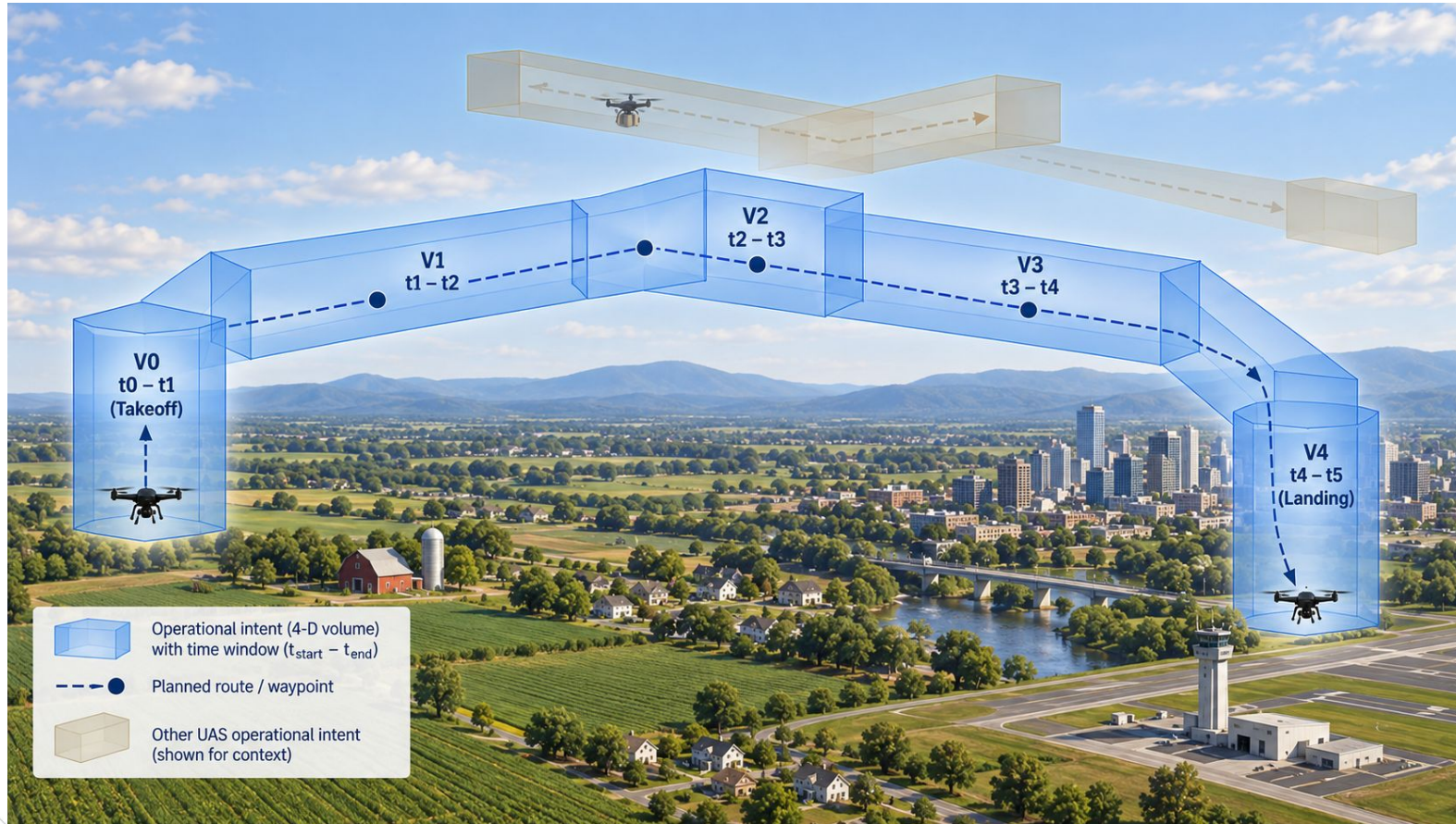
1. SADE Overview (Jane Cleland-Huang, Notre Dame) 10 mins: Start: 0.00
2. Elements of an On-Entry Decision
 - a. SAFE-CERT (Usman Gohar, Iowa State) 7 mins: Start: 0.10
 - b. Proving Ground (Robert Chris Roberts, UTEP) 8 mins: Start 0.17
 - c. Runtime Monitoring and Reputation Models (Ankit Agrawal, NorthEastern) 7 mins: Start 0.25
3. SADE Architecture and Deployment (Pedro (Tony) Alarcon Granadeno, Notre Dame) 5 mins: Start 0.32
4. SADE Simulation Environment (Doug Thain, Notre Dame) 8 mins: Start 0.37
5. Wrap-up (Jane Cleland-Huang, Notre Dame) 4 mins: Start: 0.45

The Emergent UTM Vision (Part 108)

5



Operational Intent (4D-Flight Volume)



Strategic
Deconfliction
through
preflight-
planning

Operational Intent (4D-Flight Volume)

7

Nominal / Conforming (within operational intent volume)



Off-Nominal / Nonconforming (deviated from operational intent volume, recoverable)



Off-Nominal / Contingent (deviated from operational intent volume, not recoverable)



Planned operational intent volume
(4-D, time-bounded)

---●--- Planned route / waypoint
(with time)

— Actual trajectory
(conforming)

- - -> Actual trajectory
(nonconforming - recoverable)

- - -> Actual trajectory
(contingent - not recoverable)



Off-nominal volume
(nonconforming)



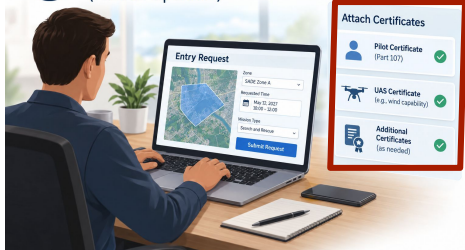
Contingency volume
(contingent)

Adapt : Dynamic Airspace Management

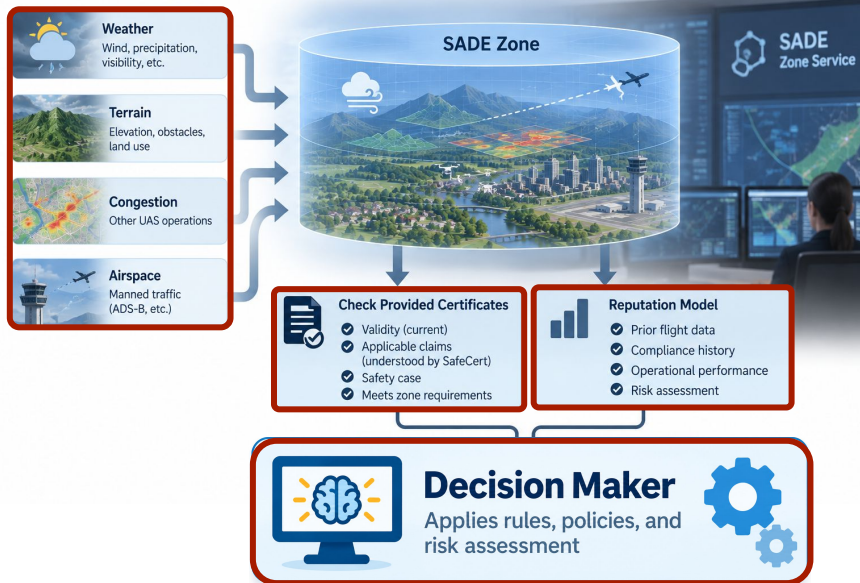
SADE: How does it work?

8

1 Submit Entry Request (Human Operator)



2 SADE Zone Evaluation (Automated + Operational Knowledge)



3

Entry Decision (SADE Zone Response)



Approve

Grant entry for requested time and zone



Deny

Do not grant entry



Request Additional Certification

Specify needed capabilities and/or evidence



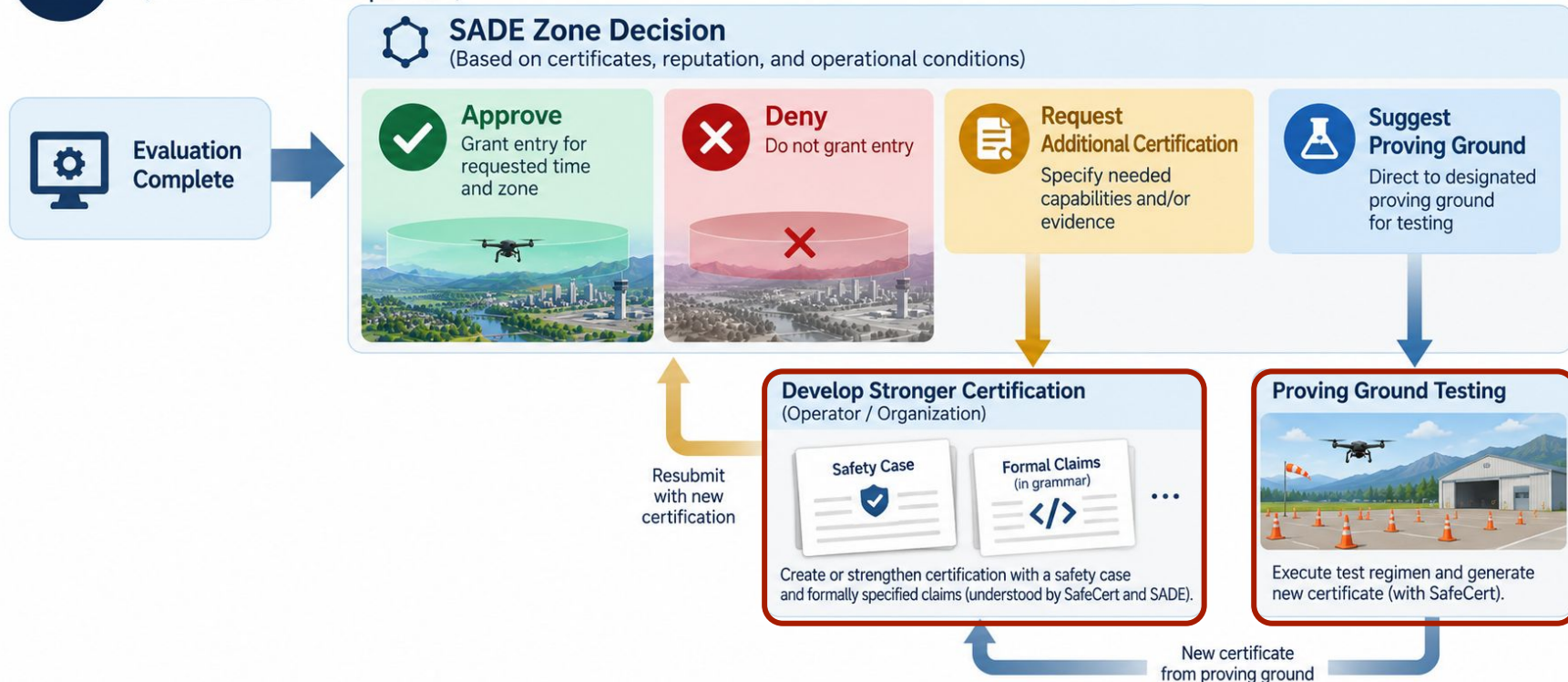
Suggest Proving Ground

Direct to designated proving ground for testing

3

Entry Decision and Next Steps

(SADE Zone Response)



Story: Fred Interacts with the UTM SADE Zone

1
Flights
1-2

Conforming Operations

Fred submits Entry Requests with valid certificates and a strong reputation. SADE admits him. Both flights remain **Conforming** and complete normally, strengthening his reputation.

2
Flight
3

Contingent Event — Altitude Excursion

SADE again admits Fred based on his certificates and reputation. During the flight, altitude is accidentally configured as absolute instead of relative (AGL). The UAV climbs to approximately 736 ft AGL, leaves its 4-D operational volume, and violates the Part 107 altitude restriction. The event is classified as **Contingent** and recorded in his reputation profile.

3
Explanation
&
Remediation

Operator Explanation and Remediation

Fred is notified of the recorded incident and given the opportunity to explain the event and describe remediation steps. He documents the root cause and corrective actions. A **human reviewer** evaluates and approves his response, and the approved explanation and remediation become part of his reputation record.

4
Next Entry
Request

Entry Approved

On Fred's next Entry Request, the SADE On-Entry Decision Maker evaluates his certificates and updated reputation. Because the prior Contingent event now includes an approved explanation and remediation, SADE **grants entry**.

Story: Alex Experiences Repeated Nonconformance in the SADE Zone

1
Flights
1-3

Repeated Nonconforming Operations

Alex is admitted to the SADE Zone. During each of three consecutive flights, geolocation problems cause the UAV to repeatedly move outside and back inside its 4D operational volume. Each flight is classified as **Nonconforming** (the UAV remains recoverable). Upon exit, SADE **notifies** Alex of the observed nonconformance and records the event in the reputation profile.

2
No
Remediation

No Explanation or Remediation Filed

After each flight, SADE gives Alex the opportunity to explain the nonconformance and provide remediation evidence. **No report is filed.** After three consecutive flights, the reputation record shows a pattern of repeated, unresolved nonconformance with the required flight corridor.

3
Next Entry
Request

Entry Denied

On Alex's next Entry Request, the SADE On-Entry Decision Maker identifies the three consecutive Nonconforming flights and the absence of remediation. **Entry is denied** because the available evidence does not provide sufficient confidence that the UAV can reliably remain within its operational volume.

4
Two Paths
Forward

Options to Restore Eligibility

SADE offers Alex two ways to regain eligibility for entry:

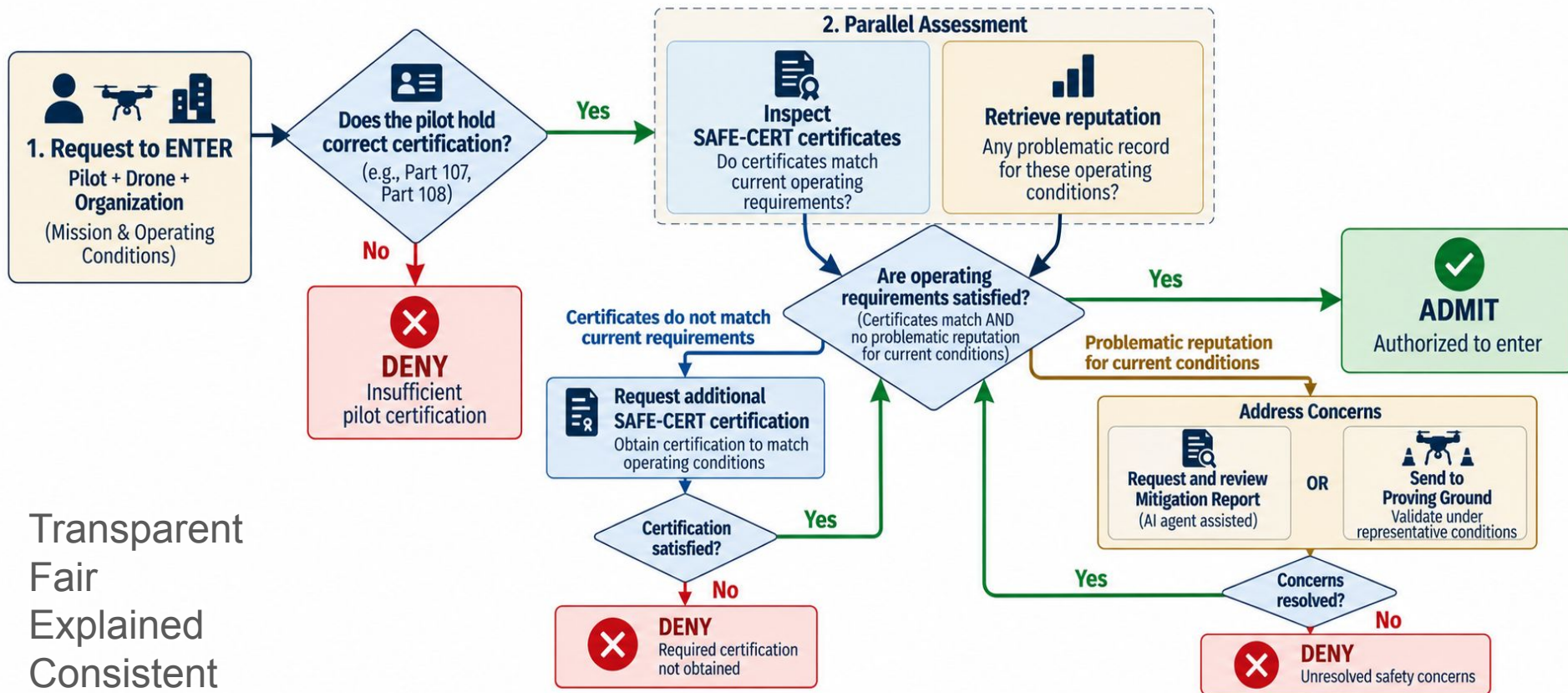
1 Proving Ground
Demonstrate the UAV's ability to reliably remain within the flight corridor through controlled testing.

OR

2 Remediation + Updated Safety Certificate
File a remediation report and provide an updated safety certificate specifically addressing the UAV's ability to remain within its flight corridor.

Decision-Making Agent

12

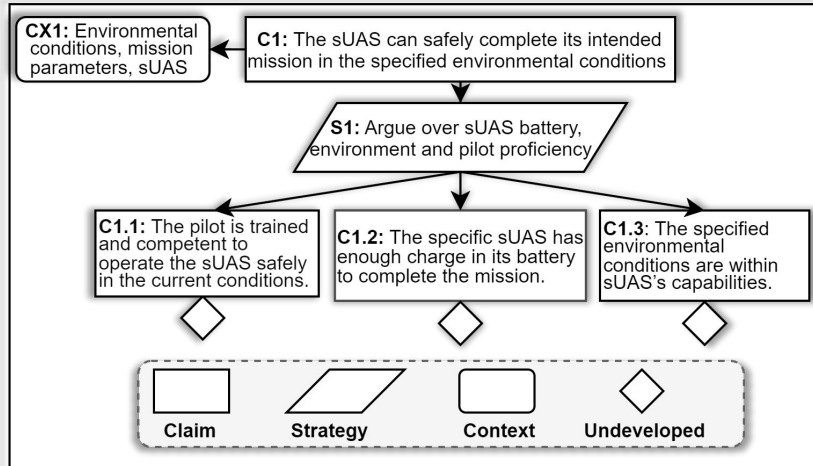


Transparent
Fair
Explained
Consistent

What is a Safety Assurance Case?

Structured argument(s), supported by evidence, intended to justify that a system is acceptably **safe** for a **specific** application in a **specific** operating **environment**

Safety Case Snippet



Widely used across safety-critical sectors through the years



Aviation



Nuclear



Railways



Autonomous Systems



Drones

Challenge?



How do we apply them to the **highly dynamic paradigm** of sUAS flights?

The Challenge of Exploding Variability in sUAS Operations

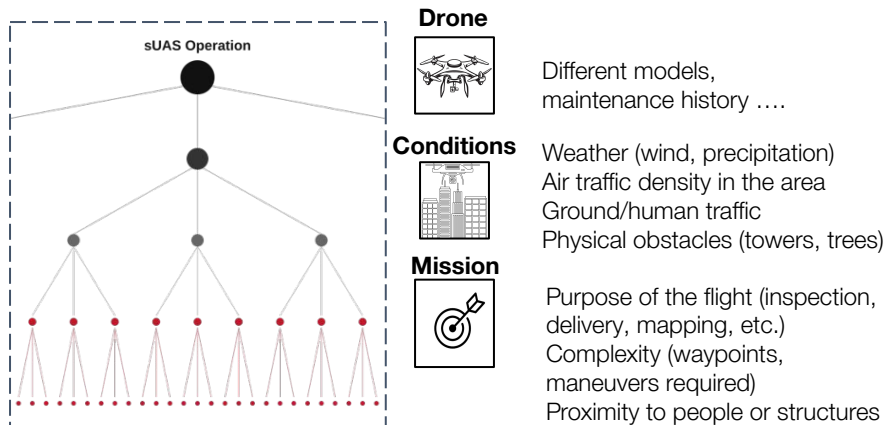
Traditional Systems

- Typically controlled and static environments (e.g., Nuclear plants, manned aircrafts)
- Procedures are largely set in advance and repeat
- Predictable scenarios and risks

sUAS Operations

- Shared, uncontrolled low-altitude airspace
- Wide range of drone makes, models, history
- Purpose and complexity of missions vary
- Very dynamic environment -> assumptions and risks require continuous reevaluation

A motivating example



Key Takeaways

Hard to Define Full Variability

A safety case with all variables will get too big and unwieldy.

Balancing Safety with Overhead

Need to balance safety with potential overhead of too many "requirements" following ALARP (As Low As Reasonably Practicable)

From FRAT to SafeCert - Safety Certificates

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Inspired by the FRAT Tool which provides a broad overview of risks in manned flights by asking questions of the pilot about their flight

Safety Certificates

Small pilot-initiated instances of evidence/logic partial safety cases to demonstrate safety

How they work?

- SADE Zone operators create safety 'certificates' i.e. partial safety cases for the relevant safety concern (e.g., windy conditions) with pilot-provided evidence.
- These indicate that a pilot is qualified and passes minimum requirements to operate safely in the SADE zone
- Ask questions of the pilots
- Certificates may include **wind, operator qualifications, BVLOS operation**, etc. (prior research, incident databases)
- Our safety certificates emulate FRAT for UAS flights in shared airspace.



Merging these individual certificates to build the overall safety case

The Flight Risk Assessment Tool (FRAT)

Runway		
Less Than 4,000 Ft	2	0
Less Than 3,000 Ft	3	0
Wet	2	0
RWY In Poor Condition	2	0

Weather		
Moderate Turbulence	2	0
Icing (Forecast/PIREP)	5	0
Thunderstorms Dep/Enroute/Dest	5	0
Crosswind Greater Than 10 Knots	2	0
Surface Winds Greater Than 20 Knots	3	0
Ceiling/Visibility - Departure Less Than 1,000 Ft/1 Sm	3	0
Ceiling/Visibility - Destination Less Than 3,000 Ft/3 Sm	3	0

Total Flight Risk Assessment	0
Risk :	<button>SAVE</button>

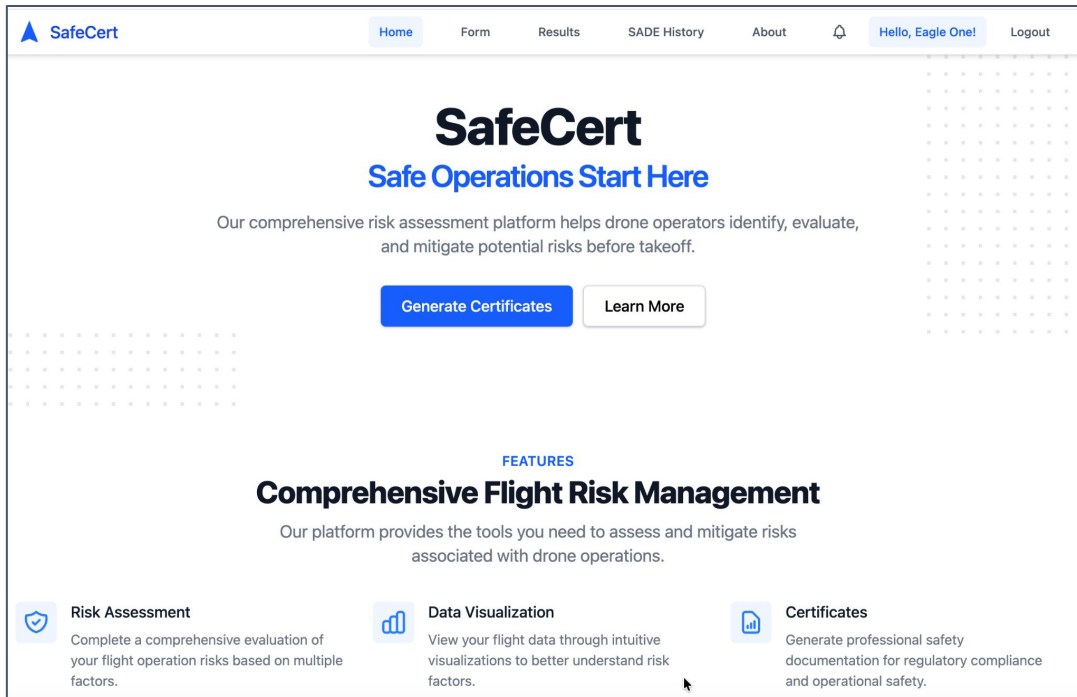
Flights

Profile

Contact

Settings

SafeCert is a web-based tool that can be used by pilots/operators to provide evidence of safety of drone flight



The screenshot shows the SafeCert web application interface. At the top is a navigation bar with the SafeCert logo, a 'Home' button, and links for 'Form', 'Results', 'SADE History', 'About', a notification bell, a user greeting 'Hello, Eagle One!', and a 'Logout' button. The main content area features the 'SafeCert' logo and the tagline 'Safe Operations Start Here'. Below this is a paragraph describing the platform's purpose: 'Our comprehensive risk assessment platform helps drone operators identify, evaluate, and mitigate potential risks before takeoff.' Two buttons, 'Generate Certificates' and 'Learn More', are provided. A 'FEATURES' section follows, titled 'Comprehensive Flight Risk Management', with a description: 'Our platform provides the tools you need to assess and mitigate risks associated with drone operations.' Three feature cards are displayed: 'Risk Assessment' (with a shield icon) describing a comprehensive evaluation of flight operation risks; 'Data Visualization' (with a bar chart icon) describing intuitive visualizations of flight data; and 'Certificates' (with a document icon) describing the generation of professional safety documentation for regulatory compliance and operational safety.

REGULATORY BASIS

- Part 107
- Proposed Part 108 NPRM
- LAANC airspace authorization
- FAA waiver process

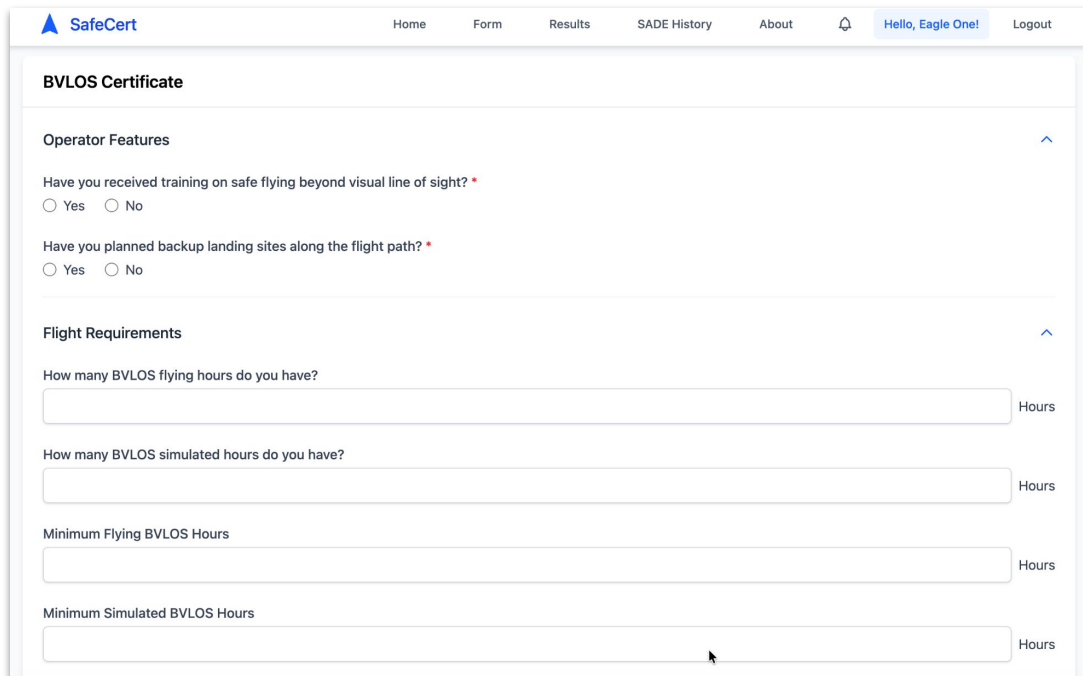
EMPIRICAL BASIS

- Incident reports
- Approved FAA waivers
- Policy orders



safecert.cs.iastate.edu (in development)

SafeCert is a web-based tool that can be used by pilots/operators to provide evidence of safety of drone flight



The screenshot displays the SafeCert web application interface. At the top, there is a navigation bar with the SafeCert logo and links for Home, Form, Results, SADE History, About, a notification bell, a user greeting 'Hello, Eagle One!', and a Logout button. The main content area is titled 'BVLOS Certificate'. It contains two sections: 'Operator Features' and 'Flight Requirements'. The 'Operator Features' section has two questions, each with 'Yes' and 'No' radio button options. The 'Flight Requirements' section has four input fields for hours, each labeled 'Hours' on the right. The questions are: 'How many BVLOS flying hours do you have?', 'How many BVLOS simulated hours do you have?', 'Minimum Flying BVLOS Hours', and 'Minimum Simulated BVLOS Hours'. A mouse cursor is visible at the bottom of the form.

BVLOS Certificate

Operator Features

Have you received training on safe flying beyond visual line of sight? *

☐ Yes ☐ No

Have you planned backup landing sites along the flight path? *

☐ Yes ☐ No

Flight Requirements

How many BVLOS flying hours do you have?

Hours

How many BVLOS simulated hours do you have?

Hours

Minimum Flying BVLOS Hours

Hours

Minimum Simulated BVLOS Hours

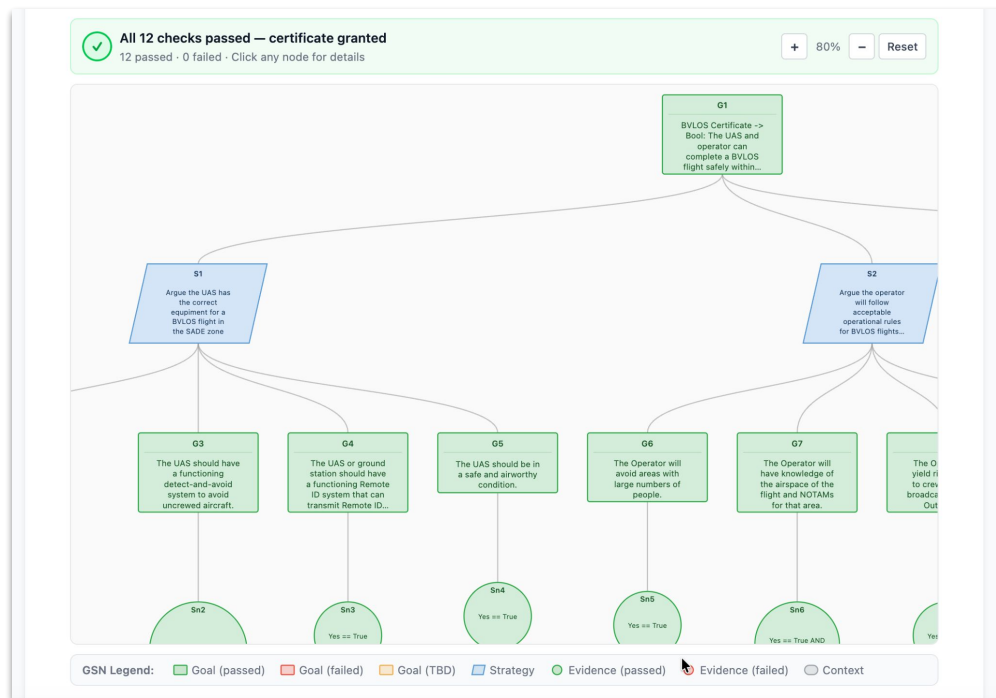
Hours

Generate Certificates

Pilot answers questions to generate certificates required by a SADE Zone. Evaluate the inputs to determine if certificate is issued



SafeCert is a web-based tool that can be used by pilots/operators to provide evidence of safety of drone flight



Generate Certificates

Pilot answers questions to generate certificates required by a SADE Zone. Evaluate the inputs to determine if certificate is issued



Introducing SafeCert

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SafeCert is a web-based tool that can be used by pilots/operators to provide evidence of safety of drone flight

SADE Request History
Every SADE entry request and how it was handled.

Request c89e3df1...

Auto-submitted Denied

Reason for denial: Mission is not permitted under current policy and conditions.

Received
Sep 8, 2026, 05:12 PM

Handled
Sep 11, 2026, 09:39 AM

Certificates submitted
Wind Certificate
Issued Jul 16, 2026 · exp Sep 24, 2026

2IVJG8QD9N ZONE-123

Request a6b2bf8f...

Auto-submitted Approved

Received
Sep 8, 2026, 05:02 PM

Handled
Sep 11, 2026, 09:39 AM

Certificates submitted
Wind Certificate
Issued Jul 16, 2026 · exp Sep 24, 2026

2IVJG8QD9N ZONE-123



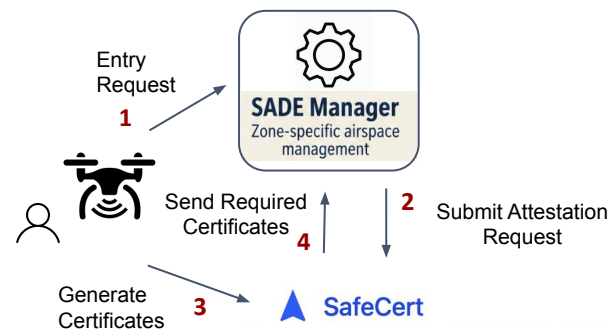
safecert.cs.iastate.edu (in development)

Generate Certificates

Pilot answers questions to generate certificates required by a SADE Zone. Evaluate the inputs to determine if certificate is issued

Handle Attestation Requests

Receives airspace-authorization requests from SADE, shows what certificates they require, and tracks their state.



SafeCert is a web-based tool that can be used by pilots/operators to provide evidence of safety of drone flight

The screenshot displays the 'Certificates' section of the SafeCert web interface. It features two tabs: 'Current' (selected) and 'Archived'. Below the tabs, there are two certificate cards. The top card is for a 'BVLOS Certificate', which is 'Passed' and 'Valid'. It includes a green status indicator, a 'Recorded on blockchain' badge (highlighted by a red arrow), and details such as 'Created Sep 11, 2026, 10:15 AM' and 'Drone 2IVJG8QD9N'. The bottom card is for a 'Configuration Certificate', which is 'Failed'. It includes a red status indicator, a list of two failed checks: 'The operator has checked the configuration of the UAS. (G2)' and 'The operator has performed preflight configuration tuning. (G4)', and details such as 'Created Sep 11, 2026, 09:26 AM'. Both cards have a 'Click to view full details' link.

Generate Certificates

Pilot answers questions to generate certificates required by a SADE Zone. Evaluate the inputs to determine if certificate is issued

Handle Attestation Requests

Receives airspace-authorization requests from SADE, shows what certificates they require, and tracks their state.

Blockchain Recording

Anchors a certificate's hash on-chain for tamper-evidence

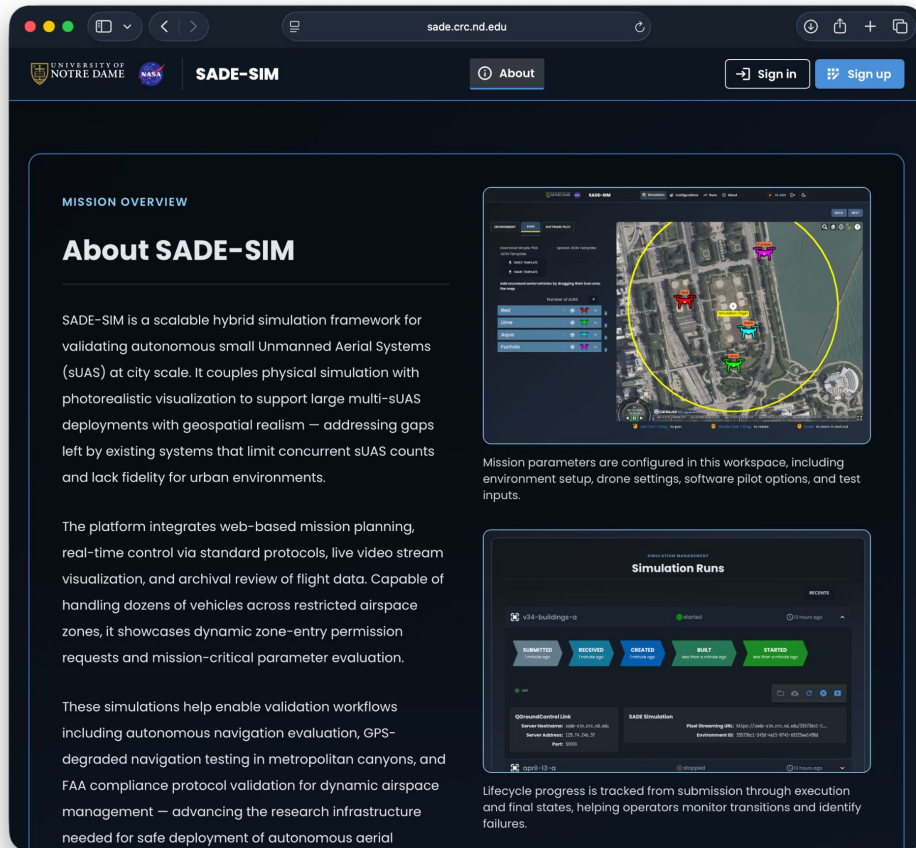
View History

Stores issued certificates, tracks validity/expiry, displays pass/fail with failure reasons



Simulation Platform Objective:

Provide a platform that enables
self-service evaluation of
custom software pilots
interacting with SADE Zones
in challenging, realistic situations.



The screenshot displays the SADE-SIM web application interface. The header includes the University of Notre Dame and NASA logos, the SADE-SIM title, and navigation links for 'About', 'Sign In', and 'Sign up'. The main content area is titled 'MISSION OVERVIEW' and 'About SADE-SIM'. It describes the platform as a scalable hybrid simulation framework for validating autonomous small Unmanned Aerial Systems (sUAS) at city scale. A sidebar on the right shows a 3D map view of a city area with various sUAS icons and a yellow circle highlighting a specific zone. Below the main text, there is a section titled 'Simulation Runs' showing a progress bar with stages: SUBMITTED, RECEIVED, CREATED, BUILT, and STARTED. The progress bar indicates that the simulation is currently in the 'STARTED' stage. The bottom of the interface shows a table of simulation runs with columns for ID, Name, Status, and Date.

MISSION OVERVIEW

About SADE-SIM

SADE-SIM is a scalable hybrid simulation framework for validating autonomous small Unmanned Aerial Systems (sUAS) at city scale. It couples physical simulation with photorealistic visualization to support large multi-sUAS deployments with geospatial realism — addressing gaps left by existing systems that limit concurrent sUAS counts and lack fidelity for urban environments.

The platform integrates web-based mission planning, real-time control via standard protocols, live video stream visualization, and archival review of flight data. Capable of handling dozens of vehicles across restricted airspace zones, it showcases dynamic zone-entry permission requests and mission-critical parameter evaluation.

These simulations help enable validation workflows including autonomous navigation evaluation, GPS-degraded navigation testing in metropolitan canyons, and FAA compliance protocol validation for dynamic airspace management — advancing the research infrastructure needed for safe deployment of autonomous aerial

Mission parameters are configured in this workspace, including environment setup, drone settings, software pilot options, and test inputs.

Simulation Runs

Progress bar showing stages: SUBMITTED, RECEIVED, CREATED, BUILT, STARTED.

Simulation ID: v3d-buildings-a

Status: started (11 hours ago)

QGroundControl Link: [Link]

SADE Simulation: [Link]

Post Processing URL: [Link]

Environment ID: [ID]

Lifecycle progress is tracked from submission through execution and final states, helping operators monitor transitions and identify failures.

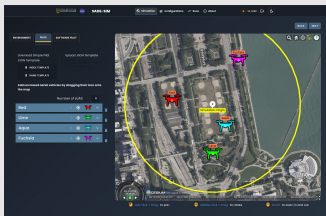
SADE Simulation Platform

Define a Scenario

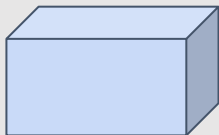
Software Pilot

```
MissionStep(  
    short_name="request_sz",  
    description="Go 1 click west  
ned=NED(north=0, east=-1 * c  
home_alt=home.altitude,  
speed=speed_mps,  
home=home,  
),
```

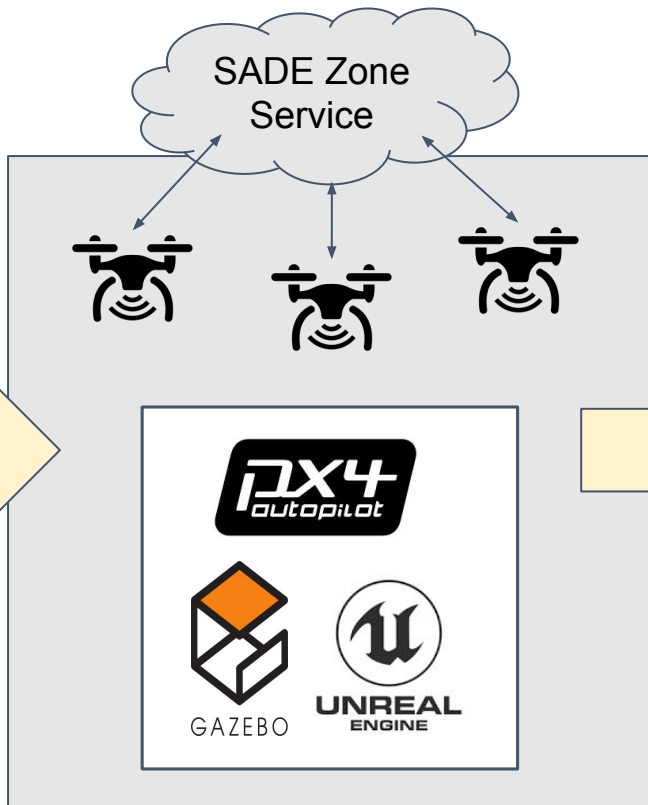
Physical Locale



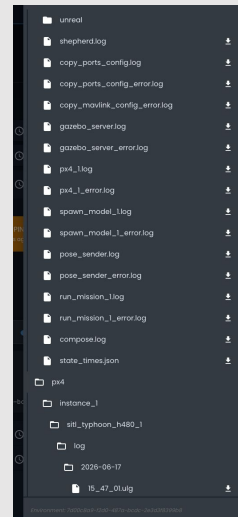
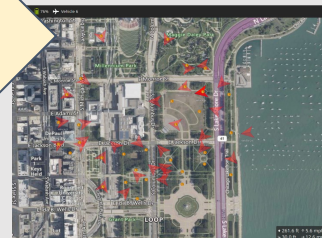
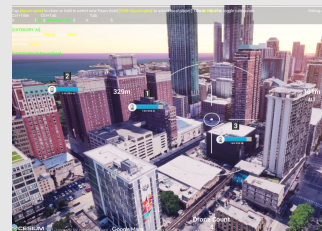
SADE Zone



Run a Simulation



Evaluate the Results



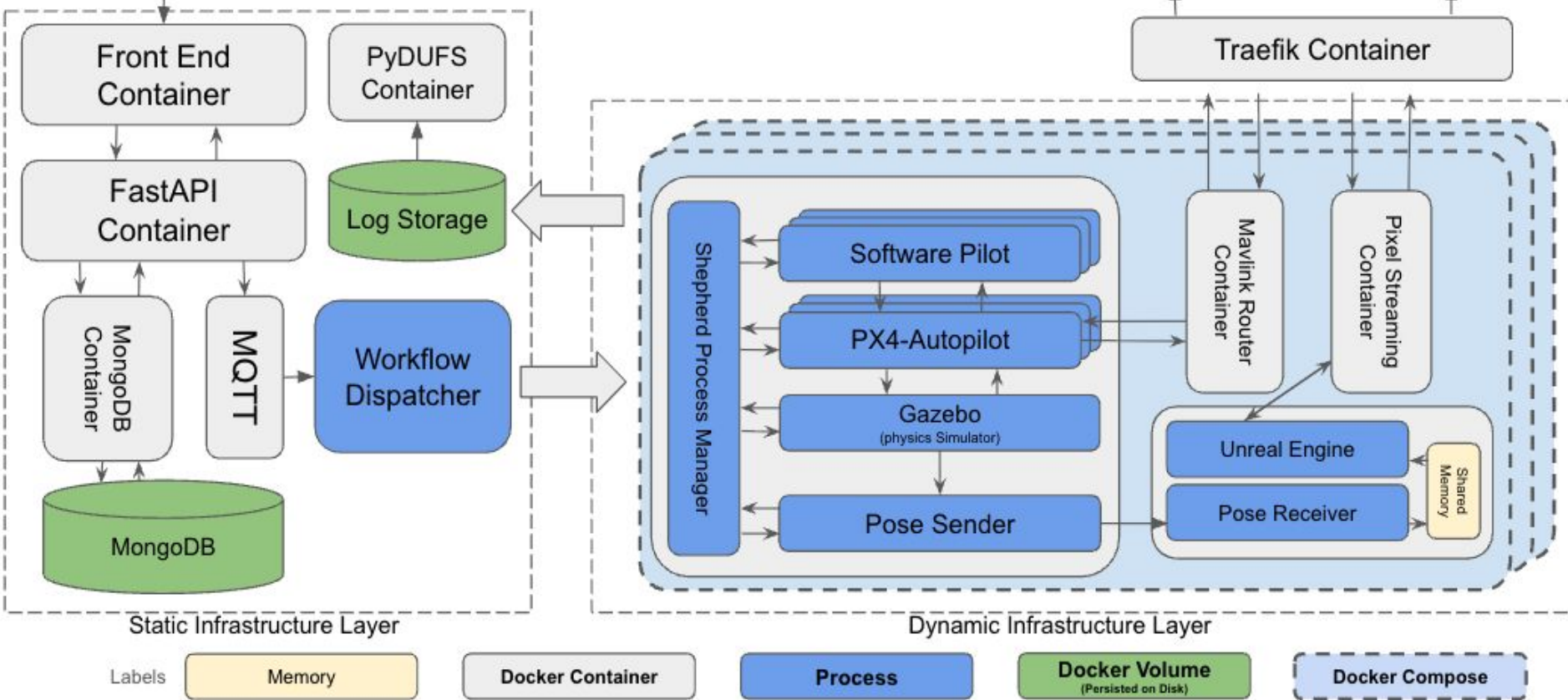
Configuration



Runtime Control



Simulation Video



BACK

NEXT

ENVIRONMENT

SUAS

SOFTWARE PILOT

SIMULATION
ORIGIN

WIND

SADE ZONE

Origin Hutchinson Field... 

Origin Latitude 41.871514

Origin Longitude -87.618865

Radius

0.3 miles



0.4828 km

Time of day 07:29:19  Left Click + Drag to pan Middle Click + Drag to rotate Scroll to zoom in and out

BACK

NEXT

ENVIRONMENT

SUAS

SOFTWARE PILOT

SIMULATION
ORIGIN

WIND

SADE ZONE

Number of SADE
Zones

- 1 +

sz-52b8

Name

sz-52b8

Ceiling (m) (sea l...

0500

Rectangle (4 vertices)

Vertex 1

Lat

41.872634490569

Lon

-87.618862513992

Vertex 2

Lat

41.872634490569

Lon

-87.616411434056

Vertex 3

Lat

41.8697306113984

Lon

-87.616411434056

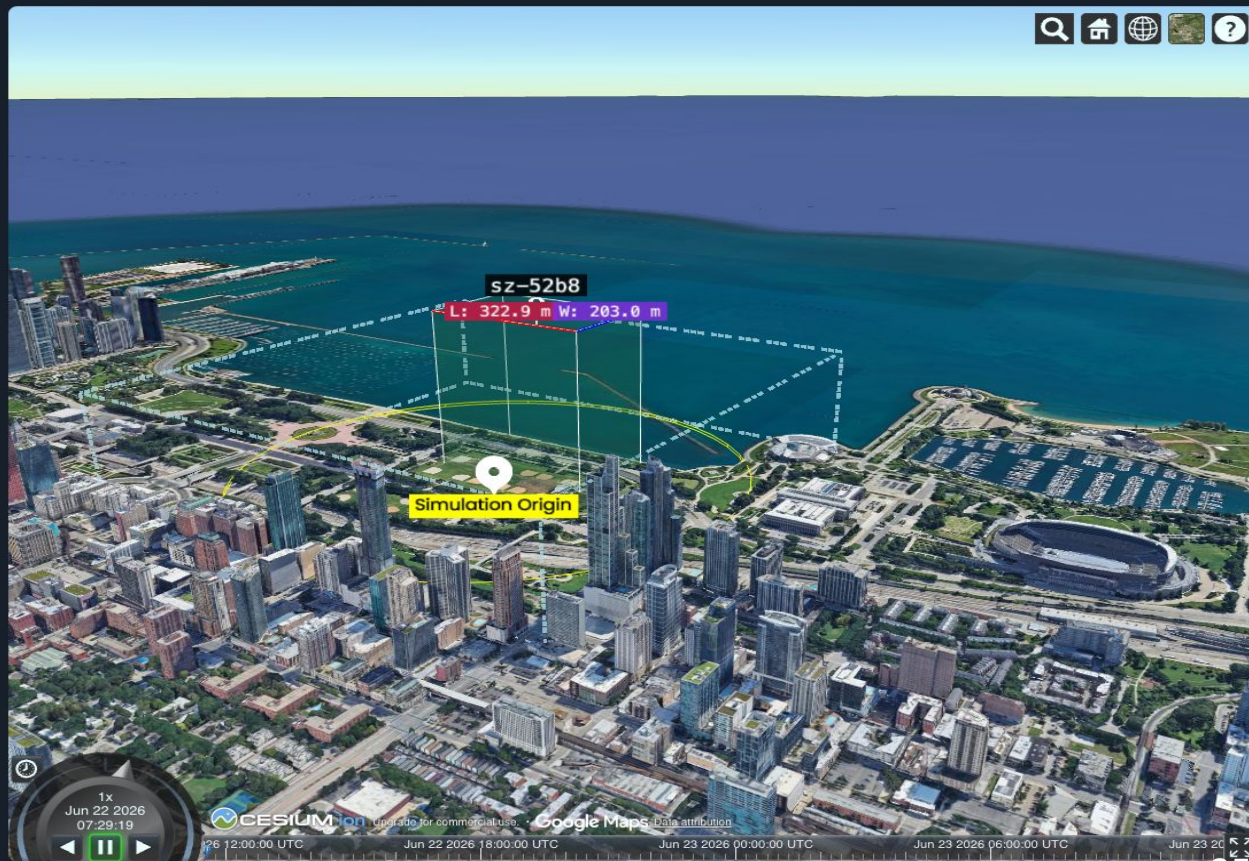
Vertex 4

Lat

41.8697306113984

Lon

-87.618862513992



BACK

NEXT

ENVIRONMENT

SUAS

SOFTWARE PILOT

Download Simple Pilot
JSON Template

INDEX TEMPLATE

NAME TEMPLATE

Upload JSON Template

UPLOAD SIMPLE PILOT
JSON TEMPLATEAdd uncrewed aerial vehicles by dragging their icon
onto the map

Number of sUAS

2

Red

1



Name

Red

Color

#ff0000

Vehicle
Type

Multi Rot...

Vehicle
Model

Parrot A...

Home Location

Latitude

41.871514

Longitude

-87.618865

Elevation (m)

145.735

Simple Pilot Configuration

Simple Pilot Params

()



BACK

FINISH

ENVIRONMENT

SUAS

SOFTWARE PILOT

BRANCHES
&
COMMITTSMERGED
PRS

Branch

chicago_case_...

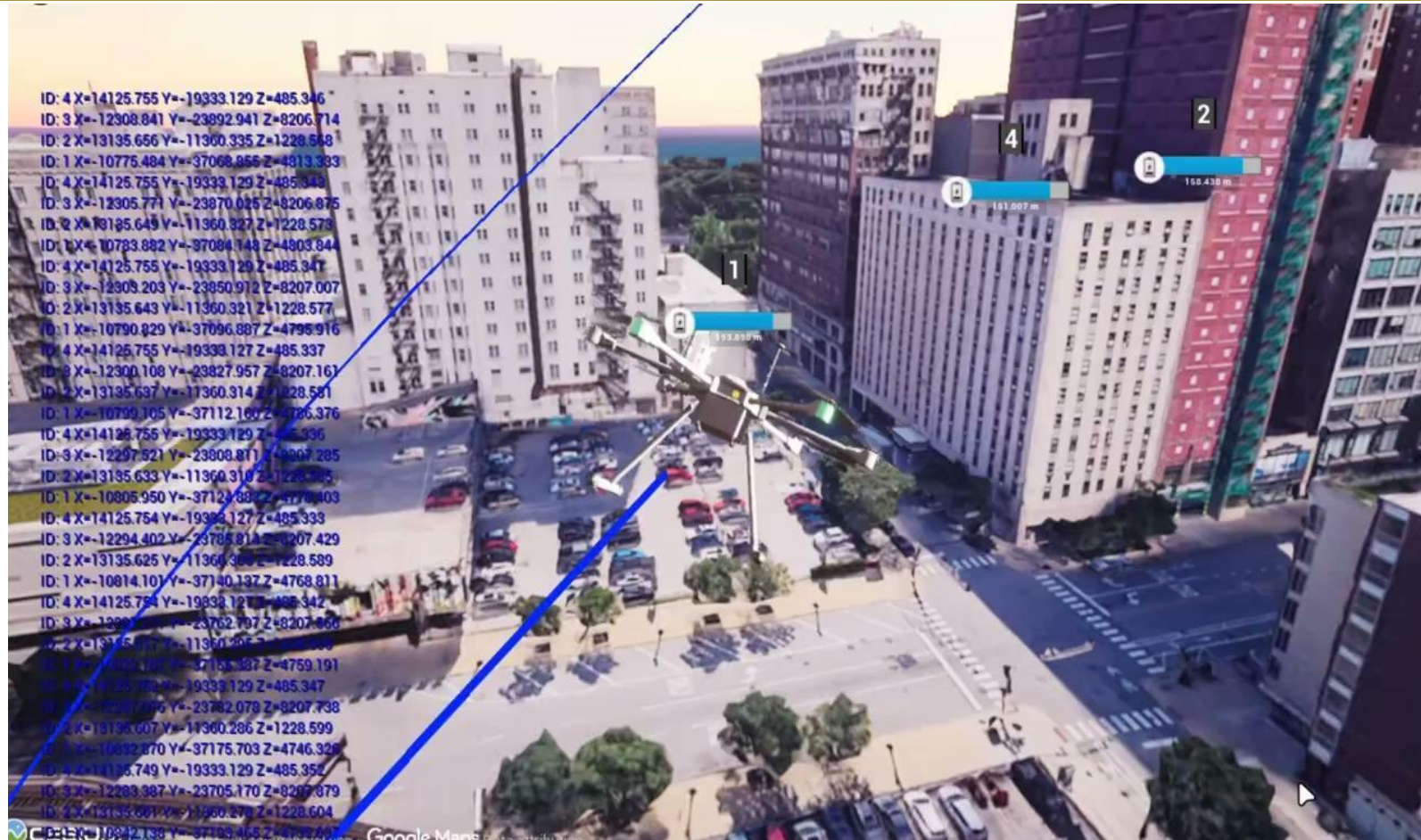
Commit ID

0caf444 • 2 months ago

Merge pull request #1 from La

[View commit on](#)[GitHub](#)

CFD Generated Wind Affects Flight Performance



Buildings and Terrain Present Real Obstacles



Shareable Simulation Configurations – Case Studies

UNIVERSITY OF NOTRE DAME

NASA

SADE-SIM

Simulation

Configurations

Runs

Case Study Runs

About

Hi, Laxminarayana!

Simulation Configurations

Upload Configuration JSON

Case Studies

Pre-configured simulation scenarios for reference (read-only)

Utah Mountains Terrain Case Study (40 Drones)

Read-only

40 2026-07-15

Yosemite National Park Search And Rescue Case Study (3 drones)

Read-only

3 2026-07-15

Indiana Restricted AirStrip Case Study (12 drones)

Read-only

12 2026-07-15

peppermint field case study (5 Drones)

Read-only

5 2026-06-17

SE4ES Case Study (Chicago 32 Drones)

Read-only

32 2026-02-19

Hutchinson Fields in Chicago

41.8796°N, -87.62187°W



0132

No wind sources configured

Case Study for the simulating more than 30 drones

VIEW

COPY AS NEW

PREVIEW JSON

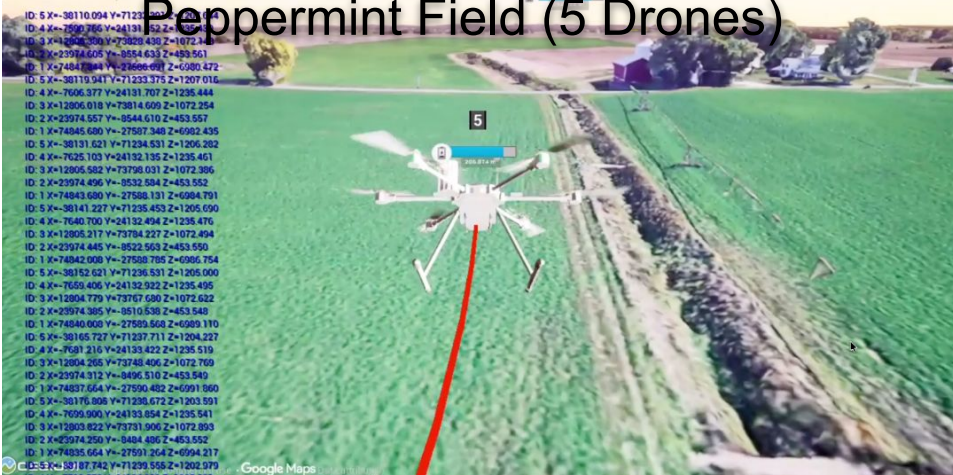
Utah Mountains (40 Drones)



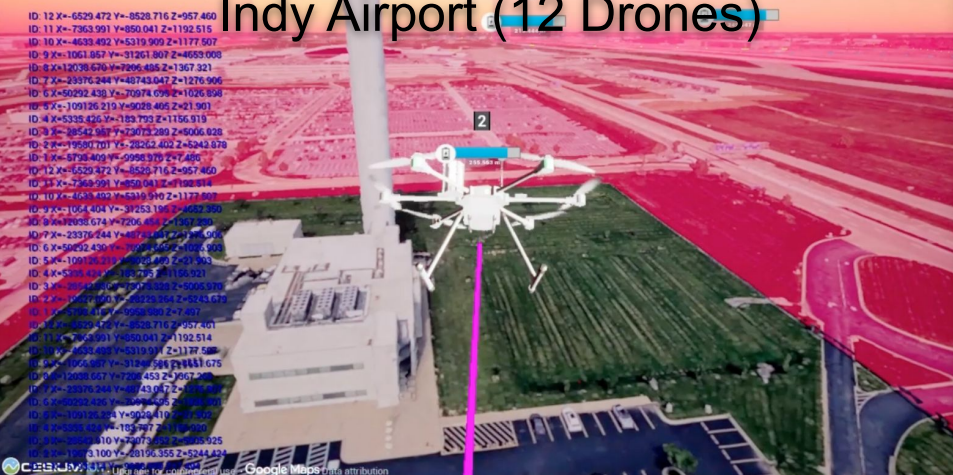
Yosemite Search and Rescue (3)



Peppermint Field (5 Drones)



Indy Airport (12 Drones)



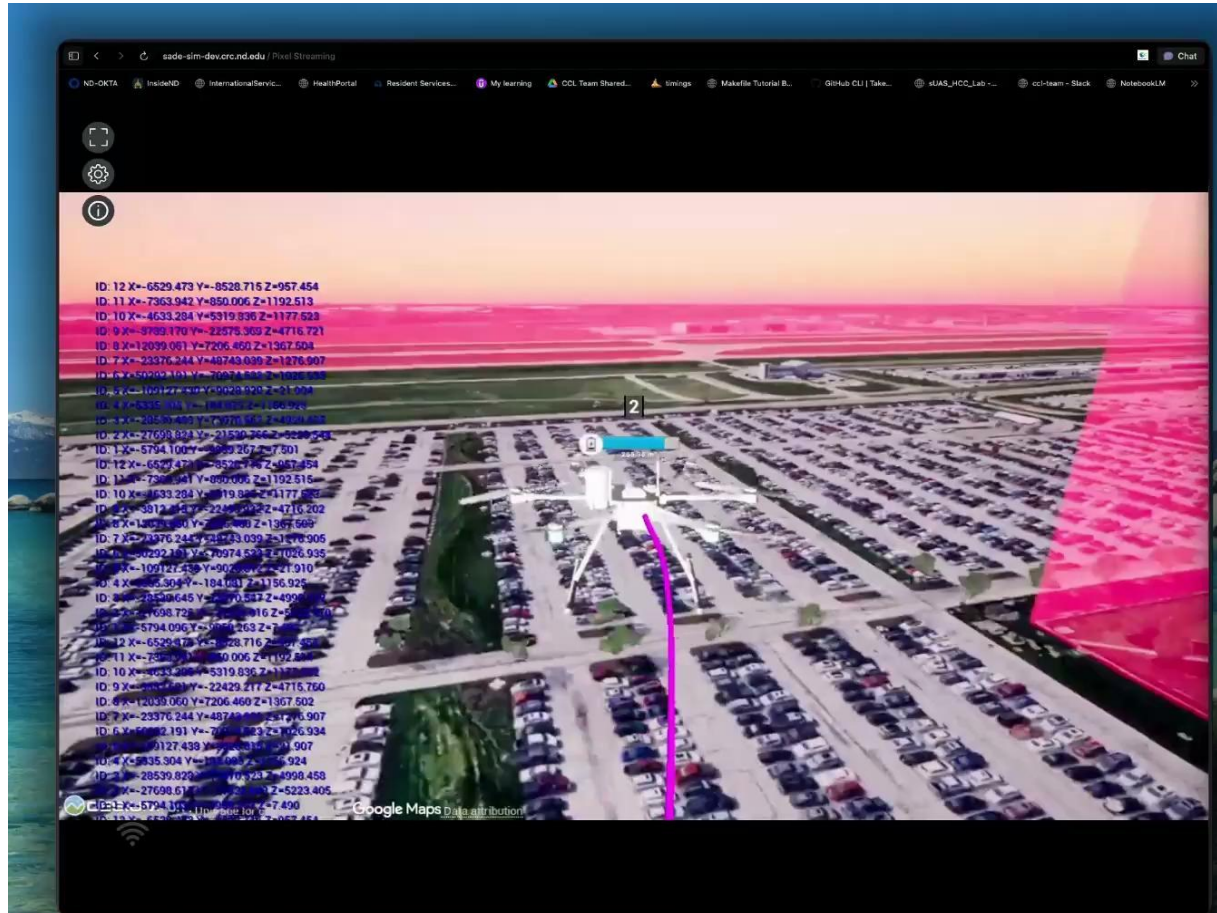
Indianapolis Airport Case Study

Drones flying civilian low-level package delivery missions in Indianapolis area.

The international airport is restricted airspace indicated by a SADE Zone.

Drone #2 is making a delivery to the airport itself: stops, requests entry, approved, and enters.

Drone #1 is passing through: stops, requests transit, denied, and navigates arounds.



Chicago Downtown Case Study

Large number of drones in downtown area with independent missions. An emergency occurs at the art museum, and a public authority declares a restricted zone.

Public Safety Drones

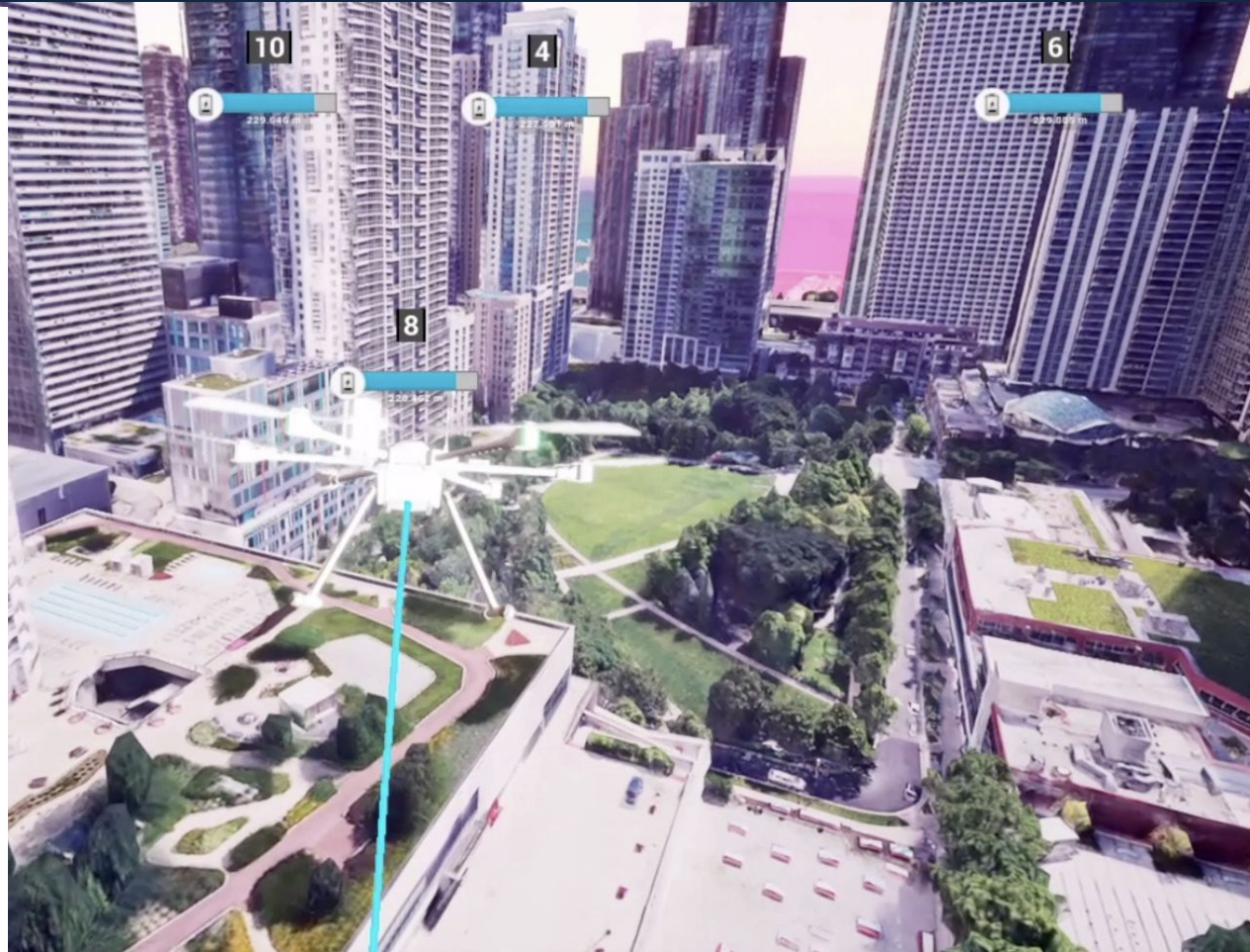
Approach, request entry, approved.

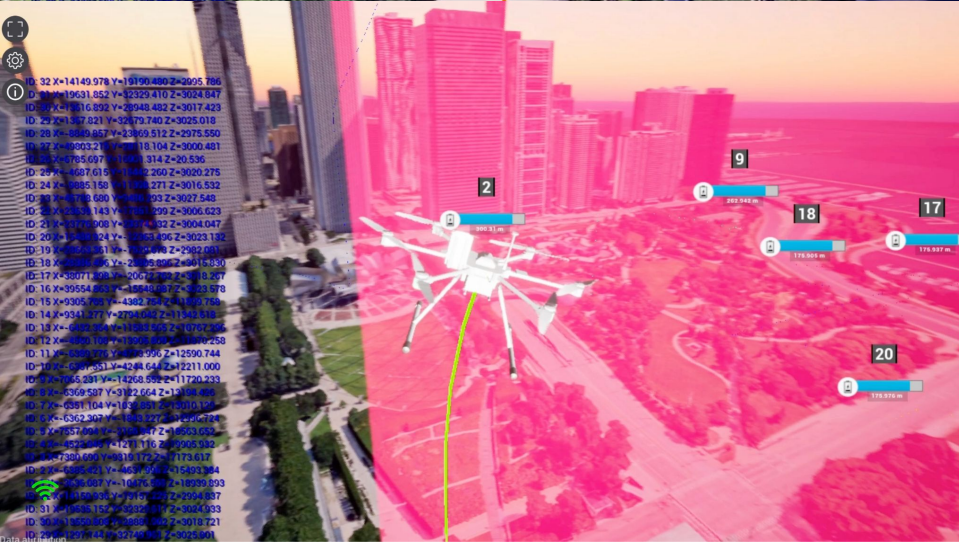
Journalism Drones:

Approach, request entry, denied, hold station near border.

Recreational Drones

Navigate independently away.





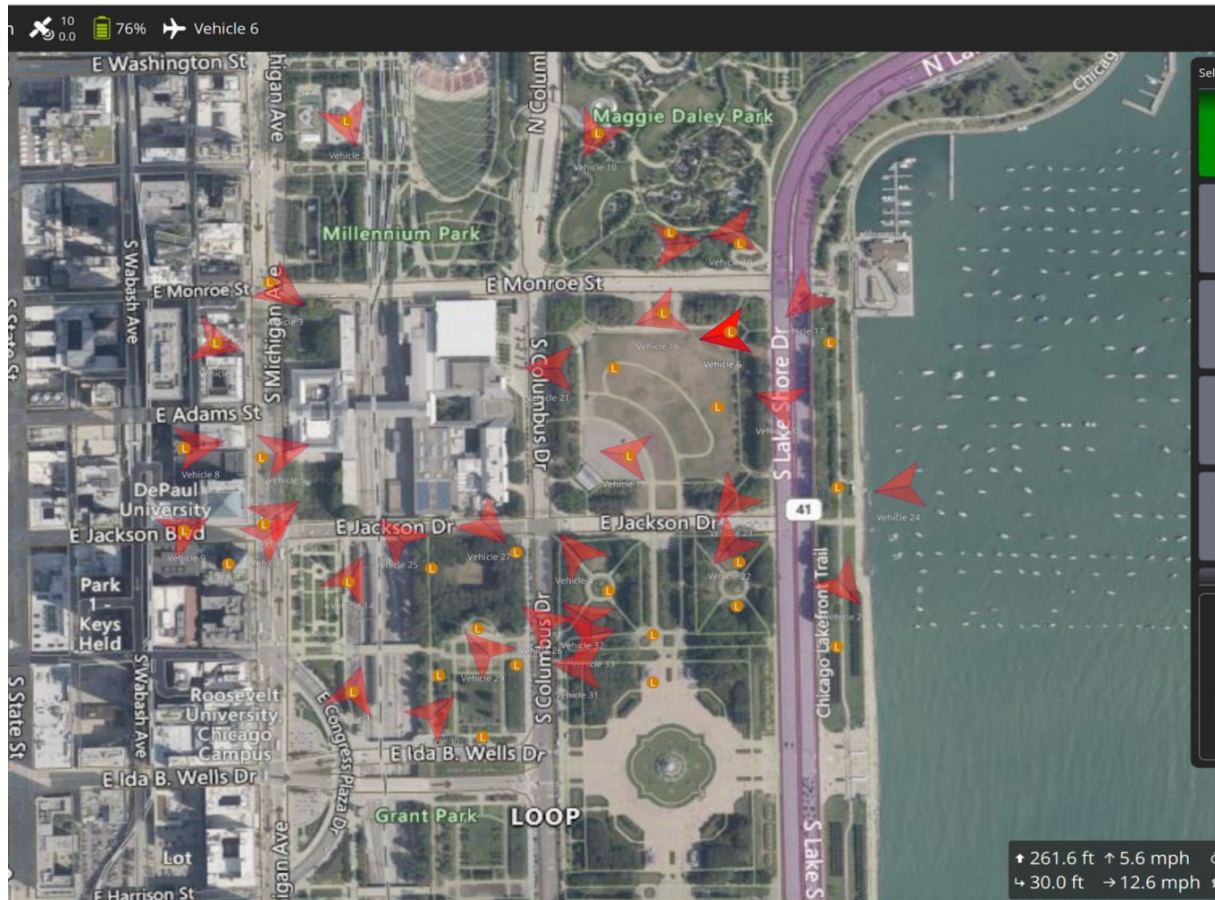
Chicago Downtown Case Study - Interoperate via MAVLink

Opens up new configurations:

- Connect an an external software pilot operating in a container elsewhere!

- Connect QGroundControl to monitor and direct a fleet of drones

- Attach RC controller to manually operate drone(s) inside simulation.





Our Inspired Flight 1200A drones are all equipped with ‘MCubes” hosting Jetson Orin NX.

Connected via Mesh Radio and Starlink to AWS.

Capable of making SADE on-entry requests and responding to:

- Approve
- Deny (go home)
- Go to Proving Ground

Preliminary tests performed with Sade Zone on-Entry



Future testing: Full On-Entry/On-Exit Life-cycle tests



Who Should Enter? Reputation, Risk, and Decision Making for UAVs in Low-Altitude Airspace

Grant #: 80NSSC23M0058 | September 15, 2026



SPEAKERS

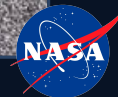
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Blockchain-Anchored Integrity for SADE Records

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Every SADE service commits a fingerprint (SHA-256 hash) of each record it issues to a shared ledger. Whoever later receives that record — from any source, at any time — can prove it is exactly what was issued.

Record producers (SADE services & agents)

SafeCert

flight safety certificate

Proving Ground

PG certificate & pilot score

SAM (Authorization Manager)

on-entry decision for an aPAR

Runtime monitoring & reports

flight record → DRP update

Hash-commitment service

Hyperledger Fabric · runs inside an Intel SGX enclave

POST /hash/store
{identifier, hash}

POST /hash/retrieve
{identifier} → hash

immutable, append-only ledger shared by SADE partners

cert-17
a3f9...

pg-42
7c02...

dec-88
e3b0...

flt-91
9d1c...

Hybrid storage (Year-2 roadmap item, now delivered): only 32-byte fingerprints go on-chain; certificates, decisions and DRPs stay off-chain — encrypted, with their owners.

Verifiers (independent record consumers)

SAM at on-entry

certificates presented with an aPAR

Post-flight & incident review

auditors / NASA re-check the records

Other zones, UTM, operators

trust records issued elsewhere

④ Recompute hash → compare

✓ match: intact ✗ mismatch: tampered

① commit

③ look up

② the record itself travels off-chain — operator app, partner API, archive, e-mail ... any channel, any time later

Write-once commitments

A fingerprint is never overwritten: storing an identifier twice returns 409 Conflict (with the existing record); an unknown identifier returns 404.

Immutable, shared ledger

Hyperledger Fabric replicated across SADE partners — no single party can rewrite history (non-repudiable).

Hardware-backed

The API and ledger run inside an Intel SGX enclave: tamper-resistant even against a compromised host.

Private & lightweight

Nothing sensitive goes on-chain — only hashes. Verification is one hash + one lookup (milliseconds).